

PAPER_571: t_neg Photon Arrival Timing via Negative Time Delay in DPM Framework

Unified Quantum Field Framework (UQFF)

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§1 Abstract

Standard light-travel-time calculations assume photons arrive at precisely $t_{\text{obs}} = r/c$. UQFF introduces a **negative-time delay** t_{neg} (PAPER_519): photons from distant shells experience a DPM-modified arrival time due to vacuum buoyancy lag, giving an adjusted observation time:

$$t_{\text{adj}} = \frac{t_{\text{obs}}}{1 + \Delta_{\text{dil}}} + t_{\text{neg}}$$

This modifies the Olbers shell brightness: photons that arrive "late" (with $t_{\text{neg}} < 0$) contribute to a different effective shell, spreading the radiance across the shell hierarchy and further damping B_{sky} .

§2 Negative Time Delay — PAPER_519

From the ShellRadiancePrototypeEquationCalculator (PAPER_519), the t_{neg} timing correction encodes DPM vacuum lag:

$$t_{\text{adj}} = \frac{t_{\text{obs}}}{1 + \Delta_{\text{dil}}} + t_{\text{neg}}$$

where $\Delta_{\text{dil}} = [\text{SSq}] \cdot (n/26)^2$ is the DPM dilation factor for shell n .

Per-shell t_{neg} distribution:

$$\Delta t_{\text{neg},n} = t_{\text{neg}} \cdot \frac{n}{26}$$

So inner shells (small n) have smaller negative delays; outer shells (large n) have the full t_{neg} .

§3 DPM-Modified Light Travel

The radial null geodesic in the DPM vacuum is modified:

$$\frac{dr}{dt}\bigg|_{\text{DPM}} = c \left(1 - \frac{\kappa_{\text{DPM}}}{[\text{SSq}]} \cdot r^{1/26}\right)$$

Integrating over shell n :

$$t_n^{\text{DPM}} = \frac{r_n}{c} + t_{\text{neg}} \cdot \frac{n}{N} + \int_0^{r_n} \frac{\kappa_{\text{DPM}}}{[\text{SSq}]} \cdot r^{1/26} \cdot dr$$

The last term gives a logarithmic correction to the classical travel time.

§4 Effect on Shell Brightness

A shell- n photon that arrives at t_{adj} instead of t_{obs} contributes to an effective redshift:

$$z_n^{\text{eff}} = z_n + \Delta z_n, \quad \Delta z_n = -H_0 \cdot |t_{\text{neg}}| \cdot \frac{n}{N}$$

Modified shell brightness:

$$B_n^{t_{\text{neg}}} = \frac{n_{\text{star}}}{L_{\text{star}}} \cdot \frac{\Delta r}{4\pi c} \cdot (1 + z_n^{\text{eff}})^4 \cdot R_{\text{Ug1},n}$$

For $t_{\text{neg}} = -1$ s (a small but non-zero delay), $\Delta z_n \approx -2.4 \times 10^{-18} \cdot n$ — negligible individually but cumulative over 26 shells.

§5 t_{neg} Gradient Effect

The gradient of B_n with respect to t_{neg} :

$$\frac{\partial B_n}{\partial t_{\text{neg}}} = -4 B_n \cdot \frac{H_0}{(1+z_n)} \cdot \frac{n}{N}$$

Summing the gradient correction over all shells:

$$\Delta B_{\text{sky}} = \sum_{n=1}^{26} B_n \cdot \Delta t_{\text{neg},n} \cdot \frac{\partial \ln B_n}{\partial t_{\text{neg}}}$$

$$= -4 H_0 t_{\text{neg}} \sum_{n=1}^{26} B_n \cdot \frac{n^2}{N(1+z_n)}$$

This provides a systematic blue/red-shift correction to the total sky brightness.

§6 DPM ProtoH Full Formula

The ProtoH formula from PAPER_519:

$$B_{\text{total}} = B_{\text{sky}}^{\text{UQFF}} + B_{\text{DPM}}^{\text{react}} \cdot P_{\text{order}} \cdot |t_{\text{neg}}|$$

In its full shell-explicit form:

$$B_{\text{total}} = \sum_{n=1}^{26} B_n \left(1 + \frac{\partial B_n}{\partial t_{\text{neg}}} \cdot |t_{\text{neg}}| \right)$$

$$= \sum_{n=1}^{26} B_n \left(1 - \frac{4 H_0 |t_{\text{neg}}| n^2}{N(1+z_n)} \right)$$

For $|t_{\text{neg}}| = 1$ s, the correction is of order 10^{-17} per shell — negligible at cosmological scales but grows with $|t_{\text{neg}}| \rightarrow t_{\text{Hubble}}$.

§7 Physical Interpretation

The t_{neg} timing effect represents **vacuum buoyancy lag**: photons from distant shells are slightly "retarded" by the DPM vacuum field, arriving later than the classical prediction. This means the universe effectively appears *younger* when observed from a UQFF perspective — reducing the effective sky brightness by delaying photon arrival from high- z shells.

The effect is coupled to the BSFG horizon blinking (PAPER_566): the $\cos(\pi t_n)$ phase in the aether metric creates a periodic t_{neg} whose average over many cycles is zero, but whose variance creates an effective line broadening in the Olbers integral.

§8 Testable Predictions

1. **Pulsar timing**: The DPM-modified geodesic predicts nanosecond deviations in pulsar arrival times as a function of n_{shell} — testable with PPTA/NANOGrav.
2. **FRB dispersion**: Fast Radio Burst dispersion measures should show a small t_{neg} excess at $z > 1$ — encoded in the DPM-modified $\text{DM} \propto \int n_e \, dt_{\text{DPM}}$.

3. **Integral effect:** The total correction $\Delta B_{\text{sky}} / B_{\text{sky}} \approx -4 H_0 |t_{\text{neg}}| \langle n^2 / (N(1+z)) \rangle$ — effectively $\sim 10^{-17}$ for $|t_{\text{neg}}| = 1$ s, but larger for $|t_{\text{neg}}| \sim t_{\text{Hubble}}$.

§9 Builds On / Addresses

Paper	Role
PAPER_519	ShellRadiancePrototypeEquationCalculator — original t_{neg} definition
PAPER_516	DPM layered shell energy — κ_{DPM} coupling
PAPER_564	DPM 26-shell Olbers (extended here with t_{neg})
PAPER_566	Gap analysis — this is Missing Extension 5

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